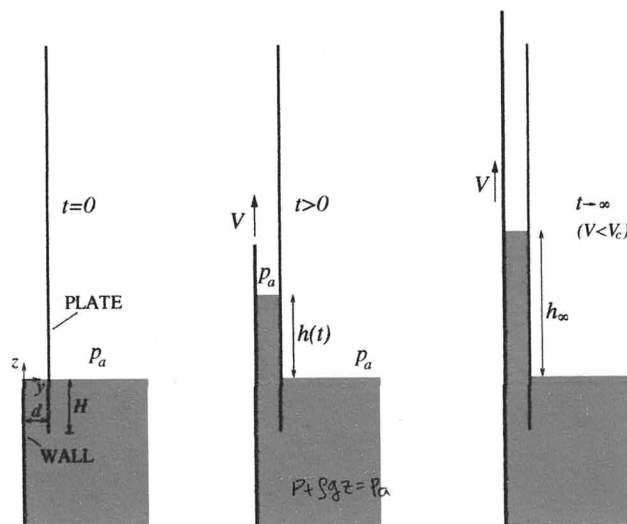


A liquid of density ρ and viscosity μ is contained in an open vessel limited laterally by a vertical wall. A thin flat plate is placed parallel to the wall with its lower end at a depth H , forming a gap of thickness d , as shown in the figure. At the initial instant of time, the vertical wall begins to move upwards with a constant velocity V , dragging the liquid into the gap, where a liquid column of height $h(t)$ rises. To solve the problem, assume that the flow in the gap is dominated by viscosity. Address the following questions in order:

1. Obtain the velocity profile across the gap $v_z(y)$.
2. Calculate the volume flow rate that enters the gap $Q(t) = \int_0^d v_z dy$ as well as the distribution of reduced pressure $P = p + \rho g z$ along the liquid column and its associated gradient $P_t = -\partial P / \partial z$. In particular, write expressions for the instantaneous values of $Q(t)$ and $P_t(t)$ in terms of $h(t)$.
3. Write the differential equation with initial condition that determines the variation with time of $h(t)$ and compute $h(t)$ by integration.
4. Investigate the solution that appears for large times. In particular, show that for values of the wall velocity V below a critical value V_c the liquid column settles at a constant height h_∞ , whereas for $V > V_c$ the column keeps rising, with a rate dh/dt that approaches a constant value $dh/dt = \dot{h}_\infty$ for large times. Give the values of V_c , h_∞ , and $\dot{h}_\infty$.
5. For $V < V_c$, compute the power needed to keep the wall in motion when the steady solution is reached.



$$1) \quad v_z = V \left(1 - \frac{y}{d}\right) + \frac{P_t}{12\mu} \frac{d^3}{d^3} \left(1 - \frac{y}{d}\right)^3$$

$$2) \quad Q = \frac{Vd}{2} + \frac{P_t d^3}{12\mu} \quad \frac{\partial P}{\partial z} = \frac{12\mu}{d^3} \left(\frac{Vd}{2} - Q\right) \quad \begin{cases} z = -H, & P = P_a \\ z = h, & P = P_a + \rho g h \end{cases}$$

$$3) \quad \frac{d}{dt} \frac{dh}{dt} = Q \Rightarrow \frac{dh}{dt} = \frac{V}{2} - \frac{\rho g h d^2}{h+H 12\mu}, \quad h(0) = 0$$

$$h - \frac{V_c H}{V - V_c} \ln \left[\frac{(V - V_c) h + V H}{V H} \right] = (V - V_c) \frac{t}{2}, \quad \text{where } V_c = \frac{\rho g d^2}{6\mu}$$

$$4) \quad \text{IF } V > V_c = \frac{\rho g d^2}{6\mu} \text{ THEN } h \rightarrow \infty \text{ AS } t \rightarrow \infty \text{ WITH A FINAL RATE } \dot{h}_\infty = \frac{V - V_c}{2}$$

$$\text{IF } V < V_c \text{ THEN } Q \rightarrow 0 \text{ AS } t \rightarrow \infty \text{ GIVING A FINAL HEIGHT } h_\infty = \frac{H V}{V_c - V}$$

$$5) \quad \vec{e}_t \cdot \vec{n} \Big|_{\text{wall}} = \mu \frac{\partial v_z}{\partial y} \Big|_{y=0} = \left(-\frac{V}{d} - \frac{\rho g h}{h+H} \frac{d}{12\mu} \right) \Big|_{h=h_\infty} = -\frac{4V\mu}{d}, \quad \dot{W} = 4\mu V^3 \frac{H}{d(V_c - V)}$$